Task 1: Install the Kickstart Package and Verify Installation

Install Kickstart Tools

- sudo dnf install -y pykickstart

Verify the installation:

- rpm -q pykickstart

Task 2: Create a Basic Kickstart Configuration File

Create the Kickstart file:

- sudo nano /var/www/html/ks.cfg

Add the following configuration:

```
#version=RHEL9
text
lang en_US.UTF-8
keyboard us
timezone Asia/Kolkata
rootpw redhat123
user --name=student --password=student123
network --bootproto=dhcp
firewall --enabled
selinux --enforcing
reboot
bootloader --location=mbr
clearpart --all --initlabel
autopart
%packages
@^minimal-environment
%end
```

Task 3: Configure an HTTP Server to Serve the Kickstart File

Install Apache:

- sudo dnf install -y httpd

Start and enable Apache:

- sudo systemctl enable --now httpd

Allow HTTP through the firewall:

- sudo firewall-cmd --permanent --add-service=http
- sudo firewall-cmd --reload

Verify the Kickstart file is available:

- curl http://<Server-IP>/ks.cfg

open in a browser:

- http://<Server-IP>/ks.cfg

Task 4: Install GNOME and Add a Post-Installation Script

Edit the Kickstart file:

- sudo nano /var/www/html/ks.cfg

Update the package section:

```
%packages
@^graphical-server-environment
%end
```

Add the post-installation section:

```
%post
mkdir -p /home/student
echo "Welcome to your automated Linux setup!" >
/home/student/welcome.txt
chown student:student /home/student/welcome.txt
%end
```

Validate the Kickstart file:

- ksvalidator /var/www/html/ks.cfg